

CodeCompass

Software Requirements Specification (IEEE 830)

Draft v1.0

1. Introduction

Purpose: Define the functional and non-functional requirements for CodeCompass, an AI-powered codebase assistant for development teams. Scope: Users add public GitHub repositories by URL. The system clones the repository temporarily, indexes and embeds supported source files, stores embeddings and metadata, deletes the local clone, and enables AI-powered chat, semantic search, documentation generation, repository insights, and GitHub-based citations.

2. Overall Description

Target users are development teams. The system uses Spring Boot, Spring AI, PostgreSQL with pgvector, React, Server-Sent Events, and temporary Git cloning. Private repositories and workspaces are out of scope for the initial release.

3. Functional Requirements

- FR-1 Authentication (JWT/OAuth).
- FR-2 Add public GitHub repository by URL.
- FR-3 Clone → Scan → Chunk → Embed → Store → Delete pipeline.
- FR-4 AI chat with streaming responses.
- FR-5 Semantic code search.
- FR-6 GitHub line-based citations using commit SHA.
- FR-7 Repository insights (frameworks, APIs, services, entities, security).
- FR-8 Documentation generation.
- FR-9 Conversation history.
- FR-10 Background indexing with IndexJob tracking.

4. Non-Functional Requirements

Performance: asynchronous indexing. Security: JWT, input validation, rate limiting. Reliability: retry failed jobs. Scalability: stateless application with PostgreSQL and pgvector. Maintainability: modular architecture and clear service boundaries.

5. Core Entities

User
Repository
RepositoryFile
CodeChunk
Chat
Message
IndexJob

6. Repository Lifecycle

User submits a GitHub URL. The system clones the repository into temporary storage, indexes supported files, generates embeddings with metadata (path, commit SHA, line numbers), stores vectors in pgvector, and removes the cloned repository.

7. Future Enhancements

Private repository support, GitHub webhooks, hybrid search, repository comparison, self-hosted LLMs, and distributed indexing.